RUBEN DARIO FLOREZ ZELA

AI Researcher · Human-Aware AI for Intelligent Vehicles · RENACYT Level V

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ORCID | Google Scholar | ResearchGate | LinkedIn | CTI Vitae | Personal Web

Citations: 171 · h-index: 5 · i10-index: 3 (Google Scholar, 2026)

RESEARCH PROFILE

AI researcher developing multimodal and deployable intelligent systems for human-aware transportation and intelligent vehicles.

My work integrates computer vision, gaze analysis, embedded AI, and multimodal deep learning to understand driver cognitive states such as attention, distraction, fatigue, and workload under real-world conditions. I design low-latency AI architectures for embedded platforms including NVIDIA Jetson devices, with applications in ADAS, transportation safety, and automated driving.

Recognized as a RENACYT Level V researcher by CONCYTEC (Peru), I serve as reviewer for international journals including *Scientific Reports (Nature Portfolio)* and *ACM Transactions on Intelligent Systems and Technology*.

My long-term research interests include human-centered AI, intelligent transportation systems, human-AI interaction, and deployable AI systems for safety-critical environments.

EDUCATION

M.Sc. in Electronic Engineering (Automation & Instrumentation) — In progress
Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

Expected Graduation: Sept. 2026

B.Sc. in Electronic Engineering — Graduated, February 2024
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

Professional License (CIP): 336757

ACADEMIC POSITIONS

Contract Lecturer – Electronic Engineering Department — August 2024 – March 2026
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Courses: Artificial Intelligence, Digital Image Processing, Robotics, Electronics Laboratory
- Focus on applied AI and embedded systems in real-world scenarios.

Teaching Assistant – Electronic Engineering Department — April 2024 – August 2024
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Assisted laboratory and practical sessions in engineering and AI-related courses.
- Supported student projects and applied learning activities in electronics and computing.

RESEARCH INTERESTS

· Human-Centered AI · Intelligent Transportation Systems · Computer Vision (RGB–NIR) · Multimodal AI · Embedded AI · Driver-State Monitoring · Automated Driving · Human-AI Interaction

SELECTED PEER-REVIEWED JOURNAL ARTICLES

[1] **Florez, R.**, Palomino-Quispe, F., Alvarez, A. B., Coaquira-Castillo, R. J., & Herrera-Levano, J. C. (2024). A Real-Time Embedded System for Driver Drowsiness Detection Based on Visual Analysis of the Eyes

and Mouth Using Convolutional Neural Network and Mouth Aspect Ratio. *Sensors*, 24(19), 6261. [35 citations]

[2] **Florez, R.**, Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., Paixão, T., & Alvarez, A. B. (2023). A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification. *Applied Sciences*, 13(13), 7849. [**93 citations**, Google Scholar 2026]

[3] Fuentes-Beingolea, J. H., Palomino-Quispe, F., Herrera-Levano, J. C., Vargas-Mateos, W., **Florez, R.**, & Alvarez, A. B. (2025). Illumination-Robust Conjunctival Image Preprocessing for Accurate Segmentation and Anemia Detection Using Deep Learning. *International Journal of Online and Biomedical Engineering (iJOE)*, 21(07), 106–124.
Coffee-Leaf Diseases and Pests Detection Based on YOLO Models. *Applied Sciences*, 15(9), 5040. [22 citations]

SELECTED PEER-REVIEWED CONFERENCE PAPERS

[1] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Control for Simplified Lumbar Spine Robotic Mechanism Motion. *2023 IEEE International Conference on Networking, Sensing and Control (ICNSC)*, Vol. 1, pp. 1–6. IEEE.

[2] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Motion Control of a Robotic Lumbar Spine Model. *Int'l Work-Conference on Bioinformatics and Biomedical Engineering (IWBBIO)*, pp. 205–216. Springer Nature.

[3] **Florez, R.**, Concha-Ramos, Y., Palomino-Quispe, F., & Coaquira-Castillo, R. J. (2022). CNN for the Detection of COVID-19 from Chest X-Ray Images. *2022 IEEE Engineering International Research Conference (EIRCON)*, pp. 1–4. IEEE.

Full publication list available / Google Scholar

PEER REVIEW SERVICE

Active reviewer for the following indexed journals:

- ACM Transactions on Intelligent Systems and Technology
- Scientific Reports (Nature Portfolio)
- Scientific Data (Nature Portfolio)
- Discover Artificial Intelligence (Springer Nature)
- Discover Applied Sciences (Springer Nature)
- Discover Internet of Things (Springer Nature)
- International Journal of Science and Technology (IJST)
- IAES International Journal of Artificial Intelligence (IJ-AI)

SELECTED RESEARCH EXPERIENCE

Driver-State Monitoring via Embedded Vision (Lead Researcher) Feb 2023 – Aug 2023
LIECAR Lab (UNSAAC)

- Developed a deployable multimodal driver monitoring system using CNN-based eye-state analysis.
- Designed ROI correction algorithms robust to illumination variability.
- Deployed real-time inference pipeline on NVIDIA Jetson Nano.

Control of a Simplified Lumbar Spine Mechanism (Research Collaborator) Feb 2023 – Jul 2023
LIECAR Lab (UNSAAC) & PAVIC Lab (UFAC, Brazil)

- Developed PID and Fuzzy Logic controllers for vertebral movement in a tendon-driven 3D-printed lumbar spine model.

Computer Vision Applied to a Low-Cost Mini Humanoid Robot (Collaborator) Mar 2021 – Apr 2022
RoboticsLab

- Implemented a computer vision guidance system enabling autonomous navigation in competition environments.

AWARDS & RECOGNITIONS

1st Place – VIII Regional Science, Technology & Innovation Fair Sep 2023
Regional Government of Cusco, OCTI & CORCYTEC | Project: COVID-19 Diagnostic System via AI

1st Place – II International Congress of Research, Innovation & Entrepreneurship Nov 2022
UNSAAC | Project: Real-Time Driver Drowsiness Detection System

INVITED TALKS & PRESENTATIONS

Drowsiness Detection System for Vehicle Drivers Using Neural Networks May 2023
VII Seminar on Industrial Electronics and Automotive Mechatronics | IESTP EPAMET

Python in Image Processing and Artificial Intelligence (Webinar) Jun 2024
Pythonistas Community

Machine Vision and AI in Robotics (Webinar) Oct 2021
Data Science Research Peru (DSRP)

Computer Vision and AI Applied to Humanoid Robots (Webinar) Oct 2021
IEEE Computer Society – UDEP Lima & Data Science Research Peru

Deep Learning for COVID-19 Diagnosis and Other Applications (Webinar) Apr 2021
IEEE RAS – UNSAAC

PROFESSIONAL MEMBERSHIPS

- IEEE Member (Region 9 – Peru Section)
- IEEE Engineering in Medicine and Biology Society (IEEE EMBS)
- IEEE Intelligent Transportation Systems Society (ITSS)
- IEEE Vehicular Technology Society (VTS)
- RENACYT Researcher Level V – CONCYTEC, Peru

TECHNICAL SKILLS

AI & Deep Learning:	TensorFlow, Keras, PyTorch, OpenCV
Computer Vision:	RGB–NIR imaging, gaze analysis, facial landmark tracking
Embedded AI:	NVIDIA Jetson Nano, Raspberry Pi, STM32, ESP32
Programming:	Python, MATLAB, C/C++

LANGUAGES

Spanish	Native
English	Professional academic reading and scientific writing